

Isabelle Laing

CONTACT	Student Email: isabelle.laing@mail.utoronto.ca Phone: +1 (416) 771-1737
EDUCATION	University of Toronto Expected Graduation in June 2027 Honours Bachelor of Science at the University of Toronto - Astronomy & Physics Specialist - Mathematics Minor
AWARDS	Best Talk Award at the Canadian Astronomical Society's Annual General Meeting, 2026 Data Science & AI Talent Showcase Poster Award , University of Toronto, 2026 Samuel Beatty Fund Travel Award , Physics Department University of Toronto, 2026 SURP Poster Award , University of Toronto, 2025 University of Toronto Admissions Scholarship , 2023
RESEARCH EXPERIENCE	SURF – Summer Undergraduate Research Program May 2026 – Present Canadian Institute for Theoretical Astrophysics (CITA), University of Toronto - Supervised by Dr. Rohan Dahale, Dr. Aviad Levis and Dr. Bart Ripperda. - Building a pipeline in Python/JAX to model polarised light curves from plasma hotspots orbiting Sgr A*. AST430 – Student Research Course September 2025 – April 2026 David A. Dunlap Department of Astronomy & Astrophysics, University of Toronto - Supervised by Dr. Renée Hložek, on the Hnauts research team Generated a full pipeline for stream simulation and investigated how different dark matter models affect the streams themselves. Student Research Internship May 2025 – August 2025 Sydney Institute for Astrophysics, University of Sydney - Supervised by Dr. Joss Bland-Hawthorn and Dr. Thorsten Tepper-Garcia. Identified and analysed clumps in galactic simulations for the Galacticos Group. SURP – Summer Undergraduate Research Program May 2024 – August 2024 David A. Dunlap Department of Astronomy & Astrophysics, University of Toronto - Supervised by Dr. Josh Speagle and Dr. Ting Li - Created a machine learning pipeline to derive stellar parameters and chemical abundances from DESI stellar spectra. - Focus on galactic archaeology, tracing the assembly history of the Milky Way halo through its surviving stars. Released a catalogue containing neural-network-derived stellar parameters and chemical abundances for over 12 million Milky Way Halo stars. - This project continued as a paid student research internship with the Department of Statistical Sciences (UofT) from September 2024 to April 2025, and then as a student research course, AST430, from May 2025 to September 2025.

PRESENTATIONS

- **Poster & Lightning Talk**, STATSTRO 2026, July 2026
 - **Talk**, Canadian Astronomical Society's Annual General Meeting, June 2026
 - **Presentation**, University of Toronto Physics Union Undergraduate Research Conference, March 2026
 - **Presentation**, Canadian Conference for Undergraduate Women & Gender Minorities in Physics, January 2026
 - **Poster**, Data Sciences Institute Talent Showcase, January 2026
 - **Poster**, Vector Foundation Models for Science Workshop, November 2025
 - **Poster**, University of Toronto Summer Undergraduate Research Program Poster Fair, August 2024 and August 2025
 - **Poster**, University of Toronto Research Opportunity Program (ROP) Poster Fair, April 2025
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PROFESSIONAL SKILLS

Languages

- Native Speaker in French and English
- B1 Spanish

Technical Skills

- Specialised in machine learning, astrostatistics, and large astrophysical datasets.
- Python (NumPy, SciPy, JAX, PyTorch, Pandas), Git, LaTeX
- Computational Astrophysics (CTA200, University of Toronto): received a final grade of 99%.

Science Communication

- Volunteer at Questacon – National Science and Technology Centre, Canberra, Australia (2022 – 2023)
 - Tutor, Freelance (2021 – 2023): Provided tutoring in Mathematics and Physics, in both French and English.
 - Tutor, Tutor Doctor ACT (Australian Capital Territory) (2021 – 2023): Provided tutoring for French, English, Mathematics and Physics.
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OUTREACH

Astronomy Student Union Executive Member (University of Toronto)

- President (2026/27 Academic Year)
- VP External Affairs (2025/26 Academic Year)
- Second Year Representative (2024/25 Academic Year)

Volunteer, ASTROTOURS (University of Toronto, 2024 – Present)

Volunteer, Astro On Tap (University of Toronto, 2024 – Present)

Volunteer, Girls in STEM Workshop Event (University of Toronto, May 2024)

Member of Corporate Relations Team, Charitable Indulgence
(University of Toronto, 2023/24 academic year)